

Chapter 1, Problem 1PP

Problem

Calculate the amount of charge represented by 6.667 billion protons.

Step-by-step solution

Step 1 of 2

The charge represented by each proton is $1.602 \times 10^{-19} \text{ C}$.

Step 2 of 2

Calculate the charge represented by 6.667 billion protons.

$$\begin{aligned} q &= (1.602 \times 10^{-19}) (6.667 \times 10^9) \\ &= 1.0681 \times 10^{-9} \text{ C} \end{aligned}$$

Therefore, the charge represented by 6.667 billion protons is $1.0681 \times 10^{-9} \text{ C}$.